

VIRTUALIZATION:

CLASS

PC

WIN 11

2 X ZION PROCESSOR
128 GB RAM
500 GB SSD - hard disk
2 TB X 2 HDD

There is a company that requires 3 apps which run on different OS.

Xoho	VOICE	SN
Win 11	Linux	MAC

Dual boot - 2 diff. OS
on same system
but only one works
at a time

* Virtualization is used to convert one physical machine into no. of logical machine.

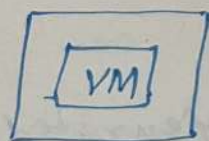
→ Instead of buying many machines for diff. apps we can use virtualization.

* We should check whether the processor supports virtualisation.

virtualisation - [Hypervisor]

- Application oriented → vmware workstation, Hyper-V, Virtual Box.
- OS oriented

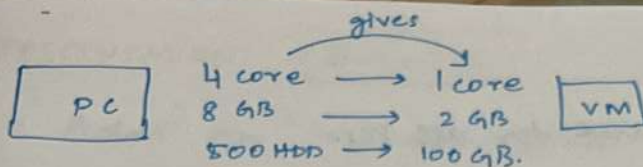
→ We have to download the applications to create a new logical machine



VM - Virtual Machine

↳ to create a PC inside PC.

PC is the physical machine (PM)



* To create more virtual machines we should have more cores & configuration accordingly.

* Even when we have 4 cores we can create multiple VMs. There are 2 ways:

① Creating multiple VM's and keeping it in OFF condition.

② Splitting the core according to the availability of no. of cores.

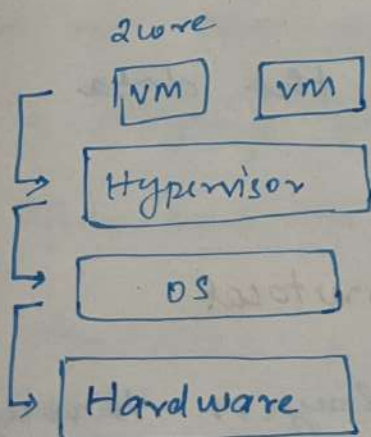
1 core → 1 VM

1 core $\begin{matrix} \nearrow \frac{1}{2} \\ \searrow \frac{1}{2} \end{matrix}$ 1 VM (It gets slowed down)
1 VM

→ Creating a VM does not make a PC slower but the usage of VMs can affect it.

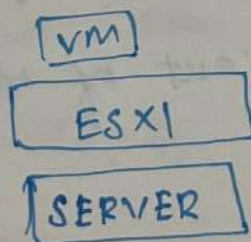
* VM is just a file to the PC. So PC cannot directly access the files of VM.

→ This is called host-based virtualization.



→ There are many layers so VM gets slow down.

→ This is Baremetal Virtualization (cost high)



→ It is an OS which acts as hypervisor